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A Additional Experimental Details

A.1 Fine-tuning ESM2

We fine-tuned the pre-trained ESM-2-650M model on the MSA of each DMS assay using LoRA (Low-Rank Adaptation) adapters to each layer. For each DMS assay, we loaded its corresponding MSA and masked 15% of the amino acids in each sequence, consistent with ESM2’s original masked language modeling objective.

To prevent overfitting, we subsample the MSA by randomly selecting up to 1000 sequences to fine-tune on. The number of fine-tuning epochs was dynamically determined based on the number of sequences used. The epoch schedule was set as follows:

- < 100 sequences: 20 epochs
- 100-299 sequences: 10 epochs
- 300-499 sequences: 5 epochs
- 500-799 sequences: 4 epochs
- ≥ 800 sequences: 3 epochs

We fine-tuned the model using the AdamW optimizer with a learning rate of 10^{-4} . We set the hyperparameters of LoRA to be the following: `r=8`, `lora_alpha=16`, `lora_dropout=0.05`, and `bias=None`.

A.2 Training SAEs

We trained a unique SAE for each of the DMS assays. For each DMS assay, we first load in the respective fine-tuned ESM2 model (Appendix A.1). We adapted code from <https://github.com/etowahadams/interpret> [14] to train on embeddings from layer 24. We dynamically set the number of training epochs based on the number of MSA sequences in each assay. The epoch schedule was set as follows:

- < 500 sequences: 1000 epochs
- 500-999 sequences: 500 epochs
- 1000-4999 sequences: 100 epochs
- ≥ 5000 sequences: 10 epochs

A.3 Fitness Extrapolation

For all tasks except regime extrapolation, we set aside 10% of the training sequences to use as a validation set and perform a grid search over regularization strengths. In regime extrapolation, we set aside 20% as validation. Since the dataset for F7YBW8_MESOW_Ding only has 166 single and double mutations, we modify the regime split to train on single, double, and triple mutations, and test on datapoints with more than three mutations. Since the DMS for SPG1_STRSG_Wu has only 4 sites with mutations, we also modify the position split to instead take 75% of amino acid positions as training positions and 25% as test.

A full summary of results can be found in Appendix B. In each fitness extrapolation table, we report the absolute value of the Spearman correlation plus the standard deviation across all nine trials.

A.4 Protein Engineering

In this section, we provide additional details on our protein engineering experimental setup. To ensure our trained MLPs can properly score the designed variants, we only design mutations at positions that are present in the DMS assays.

Feature Steering. Given the wildtype sequence, we first pass it through ESM2 to get the layer embeddings, and then pass it through the SAE encoder to get the latent representation $\mathbf{z}$. For the i^{th} predictive latent, we multiply the i^{th} row of $\mathbf{z}$ by a hyperparameter multiplier. The modified latent vector is then passed through the SAE decoder. The resulting vector is fed through the remaining